Model Comparison & Evaluation Report

Track C – Machine Learning Engineering | Encounter Total Claim Cost Prediction

Problem

Predict total claim cost of a hospital encounter using only features available at encounter start.

Data Split

Train: 2011–2019 (21,622 encounters). Test: 2020–2022 (6,269 encounters). Chronological to simulate real operational use.

Results (Test Set)

Model	Test MAE	Test RMSE	Test R^2	Test MedAE
Mean baseline	5,125	12,305	-0.002	3,598
Median baseline	3,083	12,616	-0.053	238
Ridge (linear)	4,391	11,991	0.048	2,811
Random Forest	**3,066**	11,326	0.151	1,123
Gradient Boosting	3,124	**11,279**	**0.158**	1,316

Selected production candidate: Random Forest (best MAE, competitive RMSE/R^2, interpretable feature importances).

Why the Metrics Matter

MAE is the most decision-relevant metric for operations/finance (average dollar miss). RMSE highlights the impact of the heavy right tail of costs. R^2 remains modest because procedure-level drivers of cost are intentionally excluded to prevent leakage. This is a feature of the problem definition, not a modelling failure.

Error Patterns

Highest absolute errors: rare inpatient/emergency encounters with total costs > $100k (procedure-driven). Lowest errors: ambulatory encounters. Residuals approximately centred; model does not systematically over- or under-predict on average.

Error by Encounter Class (Random Forest)

Class	n	Mean True	Mean Pred	MAE	MedAE
inpatient	131	10,732	3,829	10,427	1,824
emergency	473	5,611	4,453	4,793	1,997
urgentcare	771	6,435	5,426	4,597	1,362
wellness	326	4,827	4,430	4,156	1,615
outpatient	1,952	1,843	2,787	2,505	1,347
ambulatory	2,616	2,282	2,454	2,217	779

Leakage & Validity Checks

No future information, no STOP-derived features, no procedure table joined into features. Input validation rejects missing required columns or nulls. All unit/integration tests pass.

Next Steps for Monitoring (Pre-deployment)

1. Track MAE / RMSE by month and by ENCOUNTERCLASS.
2. Monitor prediction distribution shift (PSI or simple mean/variance checks).
3. Consider a two-stage approach: classifier for high-cost risk + regressor, or quantile regression for tail awareness.
4. Annual retrain or retrain on significant payer-mix / practice-pattern change.
5. Keep human-in-the-loop for any operational action based on high predicted cost.

Responsible Use: This is an operations cost-estimation model only. It is not a diagnostic or treatment system. No clinical validity is claimed. Synthetic data only.